Week 1 summary: sources and salient points

Satchel, a scaled-down clone of Dropbox Magic Pocket. Team: Parshv Joshi (@frectrium), @VyasGuru, @JasMehta08. Source IDs refer to the Source Log (`docs/sources/README.md`).

Sources found

Type	Count	Most important
Dropbox Tech Blog (B, primary)	15	*Inside the Magic Pocket* [B2], *Scaling to exabytes and beyond* [B1], *Pocket watch* [B3]
Conference talks (T, primary)	10	QCon SF 2022 [T8], SNIA SDC 2024 [T9], Storage Field Day 2018 [T5]
SEC filings (F, primary)	2	2018 S-1 [F1], 2018 10-K [F2]
Media (M, context only)	8	GeekWire [M5], Data Center Knowledge [M6]
Cloud pricing (P)	9	S3 pricing, ap-south-1 [P1]; Azure Retail Prices API, Central India [P5]
On-prem prices (H)	13	HDD $/TB [H2], PUE [H6], Indian colocation [H8], IP transit [H9]

Problem: why Dropbox left S3

By 2016 Dropbox held 500 PB of user data, up more than 12x from about 40 PB in 2012. File content lived on S3, while metadata ran in Dropbox's own data centers [B1]. Dropbox gave two reasons for building its own storage: **performance**, from controlling the stack end to end, and **better unit economics** at its scale [B1]. A later talk called cost the leading reason [T6]. An expiring AWS storage commitment set the migration deadline [T4].

Solution: Magic Pocket

- **Data model:** an immutable block store. Blocks are up to 4 MB and keyed by SHA-256, each stored in at least two US zones and replicated within each zone [B2].
- **Indexes and cells:** a sharded MySQL Block Index maps each block to a cell and bucket. Each cell has storage nodes (OSDs), plus a Master and Volume Managers that run repair and erasure coding outside the request path [B2].
- **Encoding:** new data is replicated (4 copies per zone), then erasure coded. The 2024 production schemes are RS(6,3) and LRC(12,2,2) [T8, T9].
- **Verification and hardware:** six continuous verifiers check durability and use over half the disk and database load [B3]. Storage runs on dense SMR servers, and the storage nodes moved from Go to Rust for memory control [B4].

Outcome

- **Migration:** over 90% of user data was served in-house by 2015-10-07, "without any major service disruptions or any loss of data" [B1]. The duplicated copies were retired in Q4 2016 [F1].
- **Cost:** infrastructure costs fell **$39.5M** in 2016 and **$35.1M** in 2017, and gross margin rose from 33% (2015) to 67% (2017) [F1]. The S-1 groups the migration with account cleanup under "Infrastructure Optimization", so these figures are not Magic Pocket savings alone.
- **Hybrid result:** AWS still serves the remaining data, delivery and European data residency [F1, T8].
- **Later savings:** SMR drives saved more than 20% over the previous storage design [B7], and the cold tier cut disk use by 25% [B5].

Inputs for the cost exercise so far

List prices	Storage per GB-month	Writes per 1,000	Internet egress per GB
S3 Standard, ap-south-1 [P1]	$0.025 down to $0.023	$0.005	$0.1093 down to $0.08
Azure Hot ZRS, Central India [P5, P6]	$0.025 down to $0.023	$0.006875	$0.12 down to $0.08

On-prem references: enterprise HDDs at $27.92–44.17 per TB in August 2026, up 15–48% since March [H2]; PUE 1.52 in Asia-Pacific [H6]; Indian colocation at Rs 25,000–60,000 per rack per month plus power [H8]; IP transit from $0.03 per Mbps per month [H9]; servers depreciated over 5–6 years [H13].

Open questions for Week 2

1. How long does Dropbox keep data replicated before erasure coding it? No source says, and it decides Satchel's parameters (#14).
2. Which workload size and year do we model: 500 PB [B1] or about 600 PB [B14] (#15)?
3. How do we update the 2019 full-server cost [H3] to 2026, given volatile drive prices [H2]?
4. Which S3 and Azure tiers match Dropbox's mostly cold, instant-read workload (#15)?